IOWA STATE UNIVERSITY

Department of Computer Science

Hopf Bifurcations as an Activation Function for Complex Recurrent Neural Networks:

A Feasibility Study

Final Oral Exam, Masters Thesis Defense

Presented by Jacob R. Pratt

Tuesday
April 16th, 2024

Program of Study Committee Members

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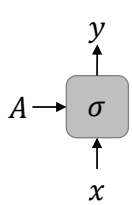
Introduction

- **Introduction**
- Related Work
- Motivations
- Proposed Method
- Experiments
  - Setup
  - Results
- Discussion
- Conclusion
- Future Works

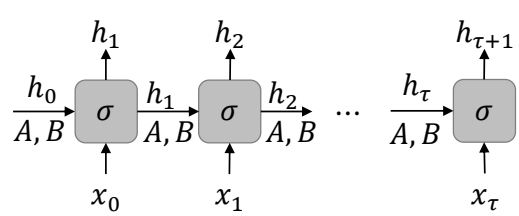
Introduction: Recurrent Neural Networks (RNN)

Feed Forward Neural Network (FNN)



$$y = \sigma(Bx)$$

Recurrent Neural Network (RNN)



$$h_{\tau+1} = \sigma(Ah_\tau + Bx_\tau)$$

Introduction: RNN Concept

- Timesteps $\tau \in \mathbb{N}$
- Hidden state h_τ acts as memory and network output
- Non-linear activation σ

Recurrent kernel $A \in \mathbb{R}^{nxn}$

Input kernel $B \in \mathbb{R}^{nxn}$

Input vector $x \in \mathbb{R}^n$

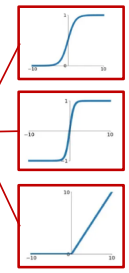
$$h'_{\tau+1} = Ah_\tau + Bx_\tau$$

Hidden State $h \in \mathbb{R}^n$

$$h_{\tau+1} = \sigma(h'_{\tau+1})$$

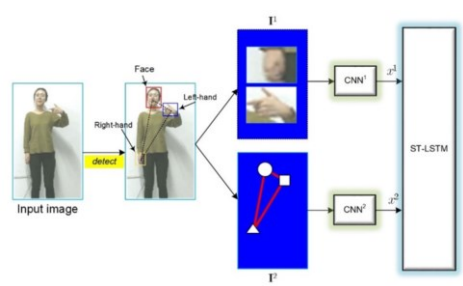
Output $h \in \mathbb{R}^n$

Non-linear activation



Introduction: Real-valued RNN Applications

- CNN-LSTM Emotion Prediction
  (Kanjo et al., 2019)
- Stock Market Prediction
  (Moghar & Hamiche, 2019); (Yan & Ouyang, 2017)
- Natural Language Processing
  (Liang, 2021); (Luo & Zhu, 2021)
- Learning Electromechanical Modeling
  (Orlovskyi, 2017)
- Continuous Sign Language Translation
  (Xiao et al., 2020)



Sign Language Feature Fusion with LSTM (Xiao et al., 2020)

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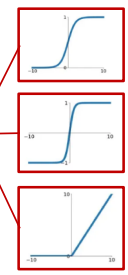


Introduction: Complex-valued RNN Concept

- Timesteps $\tau \in \mathbb{N}$
- Hidden state z_τ acts as memory and network output
- Non-linear activation σ

$z = (a + ib) = re^{i\theta}$ $\quad z \in \mathbb{C}$

$\bar{z} = (a - ib) = re^{-i\theta}$ $\quad i = \sqrt{-1}$

$\theta = arctan2(b/a)$

$r = |z| = \sqrt{a^2 + b^2}$

Recurrent kernel $A \in \mathbb{C}^{nxn}$

Input kernel $B \in \mathbb{C}^{nxn}$

Input vector $x \in \mathbb{C}^n$

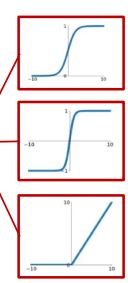
$$z'_{\tau+1} = Az_\tau + Bx_\tau$$

Hidden State $z \in \mathbb{C}^n$

$$z_{\tau+1} = \sigma(z'_{\tau+1})$$

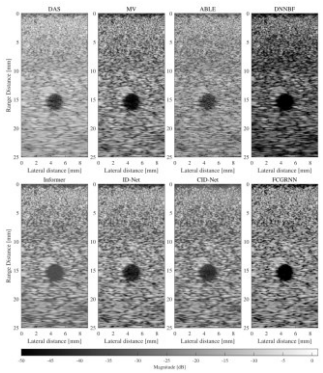
Output $z \in \mathbb{C}^n$

Non-linear activation

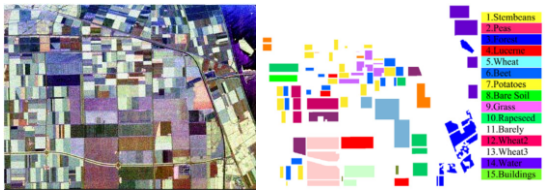


Introduction: Complex-valued RNN Applications

- Ultrasound Image Construction
  (Lei et al., 2018); (Zhang et al., 2024)



Ultrasound beamforming for cist (Lei et al., 2018)



PolSAR Image: Pauli RGB image of Flevoland dataset. (Xie et al., 2020)

- PolSAR Image Classification RCV-CNN
  (Xie et al., 2020)
- Automatic Modulation Classification
  (Zhang et al., 2024)
- Speech Enhancement Noice Filtering
  (Zhang et al., 2023)

Related Work

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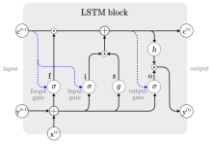
Related Works: Existing RNN Models

RNN Models

Real
- Long Short-Term Memory (LSTM). (Hochreiter et al, 1997)
- Gated Recurrent Units (GRU). (Cho et. al., 2014)
- Non-Saturating Recurrent Unit (NRU). (S. Chandar et. al., 2019)



LSTM Unit (Van Houdt et al., 2020)

Complex

Unitary
- Unitary Evolution RNN (uRNN). (M. Arjovsky et al, 2016)
- Full Capacity Unitary RNN. (S. Wisdom et. al., 2022)

$U \in \mathbb{C}^{n\times n}$ & $UU^* = I$ & U^* Conjugate Transpose of U

Gated
- Complex-Gated Recurrent Units (cgGRU). (M. Wolter et. al., 2018)
- Fully Complex-Valued Gated Recurrent Neural Network. (Lei et al., 2023)

$z_\tau, z'_\tau \in \mathbb{C}^{n\times n}$ & $z'_{\tau+1} = g_z \odot z_\tau + (1 - g_z) \odot z'_\tau$

Fourier
- Learning Long Term Dependencies via Fourier Recurrent Units. (Zhang et al., 2018)
- Resurrecting Recurrent Neural Networks for Long Sequences. (Orvieto et al., 2023)

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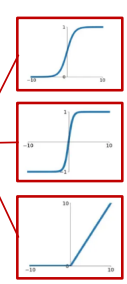
Hidden State $z \in \mathbb{C}^n$

$$z_{\tau+1} = \sigma(z'_{\tau+1})$$

Output $z \in \mathbb{C}^n$

Non-linear activation



$\sigma(\ \ )$

$tanh(\ \ )$

$relu(\ \ )$

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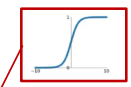
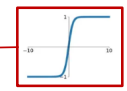
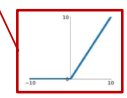


$\sigma(\ \ )$



$tanh(\ \ )$



$relu(\ \ )$

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$\bar{z} = (a - ib) = re^{-i\theta}$ $\quad i = \sqrt{-1}$

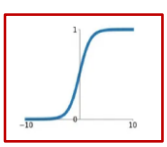
$\theta = \tan^{-1}(b/a)$

$r = |z| = \sqrt{a^2 + b^2}$

Do the common activation functions still work?

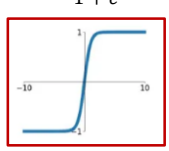
Related Works: Drawbacks with Real-valued Activation

$\sigma(z) = \dfrac{1}{1 + e^{-x}}$



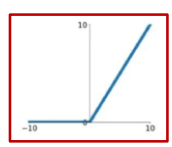
Sigmoid

$tanh(x) = \dfrac{1 - e^{-2x}}{1 + e^{-2x}}$



Hyperbolic Tangent

$relu(x) = \max(0, x)$



Rectified Linear

Example: Let $z_i = (0 + i\pi)$

Then,

$$\sigma(x) = \frac{1}{1 + e^{-(0+i\pi)}} = \frac{1}{1 + e^{-0}e^{-i\pi}} = \frac{1}{1 + (1)(\cos(-\pi) + i \cdot sin(-\pi))} = \frac{1}{1 + (1)(-1)}$$

Real-valued activation functions contain singularities in the complex-valued domain

(See: T. L. Clarke, 1990; and T. Kim and T. Adalı, 2003)

Related Works: Established Complex Activations

Transcendental	$tanh(z)$, $sinh(z)$, … , $arccos(z)$	(T.L. Clarke, 1990) (Kim & Adali, 1997)
Split	$f_{\mathbb{C}}(z) = f_{\mathbb{R}}(a) + i f_{\mathbb{R}}(b)$ $f_{\mathbb{R}}(\ldots) \in \mathbb{R} \quad z = (a + ib) \quad i = \sqrt{-1}$	(Trabelsi et al. 2018) $f_{\mathbb{R}} = ReLU$ (Nitta et al. 1997) $f_{\mathbb{R}} = \sigma$ (Popa et al. 2017) $f_{\mathbb{R}} = \sigma, ReLU$
Amplitude-Phase	$modReLU(z) = \begin{cases} (\lvert z\rvert + b)\frac{z}{\lvert z\rvert} & \text{if } \lvert z\rvert + b \geq 0 \\ 0 & \text{if } \lvert z\rvert + b < 0 \end{cases}$	(Arjovsky, 2016) (Trabelsi, 2016)
	$SigLog(z) = \frac{z}{c + \lvert z\rvert / r}$	(Georgiou & Koutsougeras, 1992)
	$CpxCard(z) = \frac{1}{2}(1 + \cos(\measuredangle z))\, z$	(Virtue, Wu, & Lustig, 2017)

Can we improve on these current methods?

Motivation

- Introduction
- Related Work
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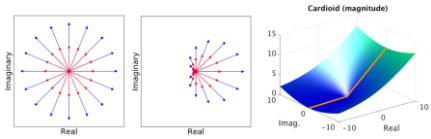
Motivation: Observations from Related Research

"We speculate that maintaining the phase of hidden states may be important for storing information across a large number of timesteps." – M. Arjovsky, A. Shah, & Y. Bengio

$$modReLU(z) = \begin{cases} (|z| + b)\dfrac{z}{|z|} & \text{if } |z| + b \geq 0 \\ 0 & \text{if } |z| + b < 0 \end{cases}$$

"We propose a new complex activation function, complex cardioid, which is sensitive to the input phase rather than input magnitude." – P. Virtue, S. Yu, & M. Lustig.

$$CpxCard(z) = \frac{1}{2}(1 + \cos(\measuredangle z))\, z$$

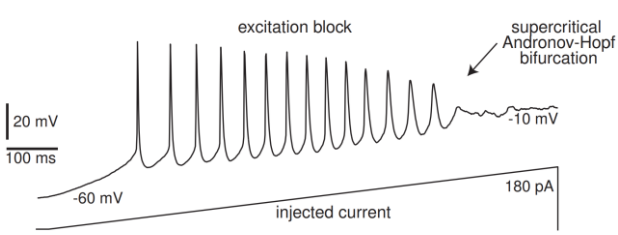


Complex Cardioid Activation Mapping (Virtue et al. 2017)

Motivation: Observations Biologicals Models

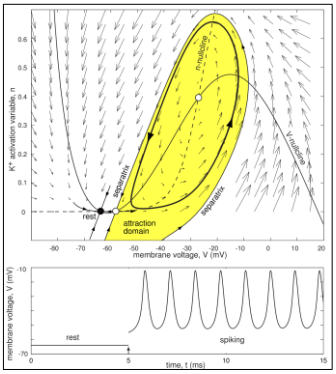
Neurons do not fire all-or-none spikes.

Neurons are resonators and prefer inputs with certain frequencies.



Biological rat neuron firing in visual cortex. (citation: A)

Cyclic Neuron Model



Cyclic firing and resting of sodium-potassium neuron model. (citation: A)

(A) "Dynamical Systems in Neuroscience: The Geometry of Excitability and Bursting." by E. M. Izhikevich, 2006

Motivation: Dynamical Systems

Dynamical System

$z_{t_j} \in \mathbb{C}^n$

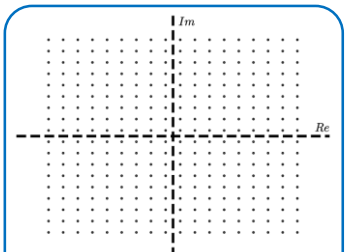
$t_0, t_1 \in \mathbb{R}$

Final State $z \in \mathbb{C}^n$

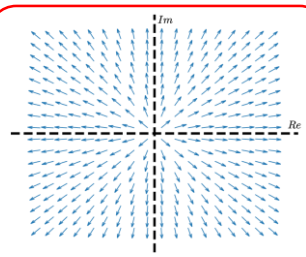
Initial State $z \in \mathbb{C}^n$

Evolution Function $f \in \mathbb{C}$

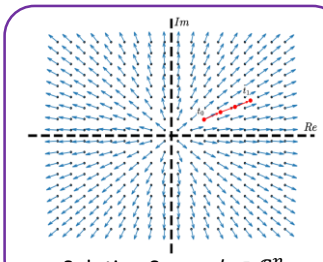
$$z_{t_1} = \phi(z_{t_0}, f, t_0, t_1) = z_{t_0} + \int_{t_0}^{t_1} f(z,t)\, dt \qquad \dot{z} = f(z,t)$$



State Space $z_t \in \mathbb{C}^n, \forall t \in \mathbb{R}$



Space Evolution $f \in \mathbb{C}^n$



Solution Space $\phi \in \mathbb{C}^n$

Motivation: Dynamical Systems with Limit Cycles

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Hopf Normal Form

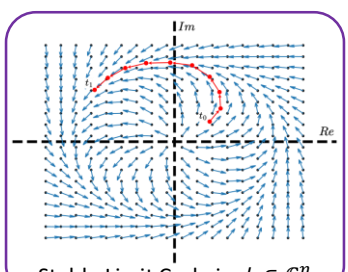
2nd Order ODE $\quad z \in \mathbb{C}^n$

$$\dot{z} = (\alpha + \beta|z|^2)z$$

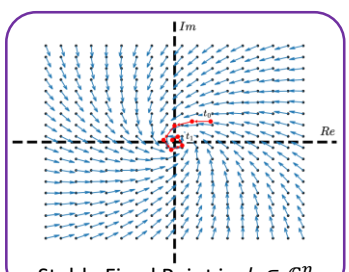
$$\alpha = (a + ib)$$

$$\beta = (c + id)$$

(Wiggins, 1990)



Stable Limit Cycle in $\phi \in \mathbb{C}^n$



Stable Fixed Point in $\phi \in \mathbb{C}^n$

Motivation: Dynamical Systems with Limit Cycles

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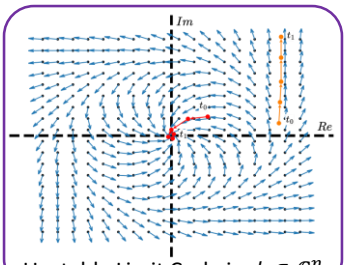
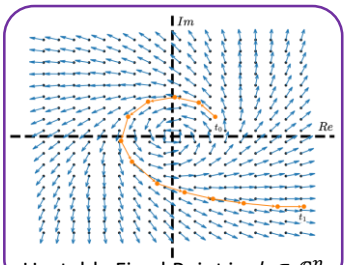


Unstable Limit Cycle in $\phi \in \mathbb{C}^n$



Unstable Fixed Point in $\phi \in \mathbb{C}^n$

Motivation: Hypothesis

Questions 1

Can a limit cycle be applied as a complex activation function?

Questions 2

What limit cycle parameters allow control and suitable output performance?

Questions 3

How does a limit cycle activation compare to similar activation functions?

Proposed Method

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Proposed Method : Dynamical System to Hopf Activation

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Hopf Activation Requirements

1) Ensure Stability

2) Ensure Complementary Phases

3) Solution Method

Proposed Method : Hopf Activation

Hopf Activation

$\gamma_{t_j} \in \mathbb{C}^n$

$t_0, t_1 \in \mathbb{R}$

Final State $\gamma \in \mathbb{C}^n$

Initial State $\gamma \in \mathbb{C}^n$

Input constants $\alpha, \beta \in \mathbb{C}^n$

Evolution Function $f \in \mathbb{C}$

$$\gamma_{t_1} = \eta(\gamma_{t_0}, \alpha, \beta) = \gamma_{t_0} + \int_0^{2\pi} f(\gamma, t)\, dt \qquad f = (\alpha + \beta|\gamma|^2)\gamma$$

Hopf Activation Requirements

1) Ensure Stability

2) Ensure complementary phases

3) Solution Method

Proposed Method : Hopf Activation

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$t_0, t_1 \in \mathbb{R}$

Final State $\gamma \in \mathbb{C}^n$

Initial State $\gamma \in \mathbb{C}^n$

Input constants $\alpha, \beta \in \mathbb{C}^n$

Evolution Function $f \in \mathbb{C}$

$$\gamma_{t_1} = \eta(\gamma_{t_0}, \alpha, \beta) = \gamma_{t_0} + \int_0^{2\pi} f(\gamma, t)\, dt \qquad f = (\alpha + \beta|\gamma|^2)\gamma$$

Hopf Activation Requirements

1) Ensure Stability $\quad \alpha = (a + ib) \quad \beta = (c + id)$

		$c < 0$		$c > 0$	
		Limit-cycle	Fixed point	Limit-cycle	Fixed point
$a > 0$	$\mu > 0$	Stable	Unstable		Unstable
	$\mu < 0$		Stable	Unstable	Stable
$a < 0$	$\mu > 0$		Stable	Unstable	Stable
	$\mu < 0$	Stable	Unstable		Unstable

Gathered from (Wiggins, 1990)

Proposed Method : Hopf Activation

Hopf Activation

$\gamma_{t_j} \in \mathbb{C}^n$

$t_0, t_1 \in \mathbb{R}$

Final State $\gamma \in \mathbb{C}^n$

Initial State $\gamma \in \mathbb{C}^n$

Input constants $\alpha, \beta \in \mathbb{C}^n$

Evolution Function $f \in \mathbb{C}$

$$\gamma_{t_1} = \eta(\gamma_{t_0}, \alpha, \beta) = \gamma_{t_0} + \int_0^{2\pi} f(\gamma, t)\, dt \qquad f = (\alpha + \beta|\gamma|^2)\gamma$$

Hopf Activation Requirements

1) Ensure Stability $\quad \alpha = (a + ib) \quad \beta = (c + id) \quad$ Require: $c < 0$

		$c < 0$		[redacted]
		Limit-cycle	Fixed point	
$a > 0$	$\mu > 0$	Stable	Unstable	
	$\mu < 0$		Stable	
$a < 0$	$\mu > 0$		Stable	
	$\mu < 0$	Stable	Unstable	

Gathered from (Wiggins, 1990)

Proposed Method : Hopf Activation

Hopf Activation

$\gamma_{t_j} \in \mathbb{C}^n$

$t_0, t_1 \in \mathbb{R}$

Final State $\gamma \in \mathbb{C}^n$

Initial State $\gamma \in \mathbb{C}^n$

Input constants $\alpha, \beta \in \mathbb{C}^n$

Evolution Function $f \in \mathbb{C}$

$$\gamma_{t_1} = \eta(\gamma_{t_0}, \alpha, \beta) = \gamma_{t_0} + \int_0^{2\pi} f(\gamma, t)\, dt \qquad f = (\alpha + \beta|\gamma|^2)\gamma$$

Hopf Activation Requirements

1) Ensure Stability $\quad \alpha = (a + ib) \quad \beta = (c + id) \quad$ Require: $c < 0$

Since:

$t_0, t_1 > 0$

$\Rightarrow \mu > 0$

		$c < 0$		[redacted]
		Limit-cycle	Fixed point	
$a > 0$	$\mu > 0$	Stable	Unstable	
	[redacted]			
$a < 0$	$\mu > 0$		Stable	
	[redacted]			

Gathered from (Wiggins, 1990)

Motivation: Hopf Activation

Hopf Activation

$\gamma_{t_j} \in \mathbb{C}^n$

$t_0, t_1 \in \mathbb{R}$

Final State $\gamma \in \mathbb{C}^n$

Initial State $\gamma \in \mathbb{C}^n$

Input constants $\alpha, \beta \in \mathbb{C}^n$

Evolution Function $f \in \mathbb{C}$

$$\gamma_{t_1} = \eta(\gamma_{t_0}, \alpha, \beta) = \gamma_{t_0} + \int_0^{2\pi} f(\gamma, t)\, dt \qquad f = (\alpha + \beta|\gamma|^2)\gamma$$

Hopf Activation Requirements

$\alpha = (a + ib) \qquad \beta = (c + id)$

1) Ensure Stability — Require: $c < 0$

2) Ensure complementary phases

3) Solution Method

Proposed Method : Hopf Activation

Hopf Activation

$\gamma_{t_j} \in \mathbb{C}^n$

$t_0, t_1 \in \mathbb{R}$

Final State $\gamma \in \mathbb{C}^n$

Initial State $\gamma \in \mathbb{C}^n$

Input constants $\alpha, \beta \in \mathbb{C}^n$

Evolution Function $f \in \mathbb{C}$

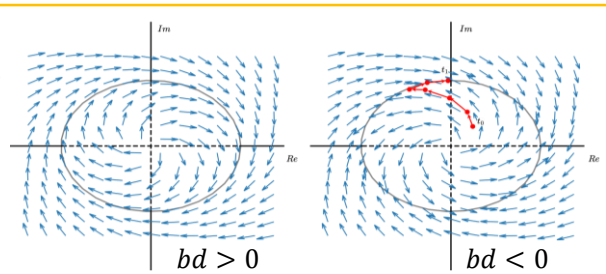
$$\gamma_{t_1} = \eta(\gamma_{t_0}, \alpha, \beta) = \gamma_{t_0} + \int_0^{2\pi} f(\gamma, t)\, dt \qquad f = (\alpha + \beta|\gamma|^2)\gamma$$

Hopf Activation Requirements

2) Ensure complementary phases

$\alpha = (a + ib) \quad \beta = (c + id)$

Require: $bd > 0$



$bd > 0$ $\qquad$ $bd < 0$

Proposed Method : Hopf Activation

Hopf Activation

$\gamma_{t_j} \in \mathbb{C}^n$

$t_0, t_1 \in \mathbb{R}$

Final State $\gamma \in \mathbb{C}^n$

Initial State $\gamma \in \mathbb{C}^n$

Input constants $\alpha, \beta \in \mathbb{C}^n$

Evolution Function $f \in \mathbb{C}$

$$\gamma_{t_1} = \eta(\gamma_{t_0}, \alpha, \beta) = \gamma_{t_0} + \int_0^{2\pi} f(\gamma, t)\, dt \qquad f = (\alpha + \beta|\gamma|^2)\gamma$$

Hopf Activation Requirements

$\alpha = (a + ib) \qquad \beta = (c + id)$

1) Ensure Stability — Require: $c < 0$

2) Ensure complementary phases — Require: $bd > 0$

3) Solution Method

Proposed Method : Hopf Activation

Hopf Activation

$\gamma_{t_j} \in \mathbb{C}^n$

$t_0, t_1 \in \mathbb{R}$

Final State $\gamma \in \mathbb{C}^n$

Initial State $\gamma \in \mathbb{C}^n$

Input constants $\alpha, \beta \in \mathbb{C}^n$

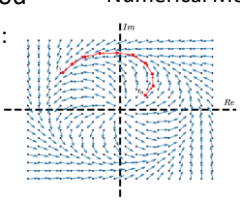
Evolution Function $f \in \mathbb{C}$

$$\gamma_{t_1} = \eta(\gamma_{t_0}, \alpha, \beta) = \gamma_{t_0} + \int_0^{2\pi} f(\gamma, t)\, dt \qquad f = (\alpha + \beta|\gamma|^2)\gamma$$

Hopf Activation Requirements

3) Solution Method — Numerical Methods give "approximate" solutions. (Chen et. al., 2018)

Forward Operation:

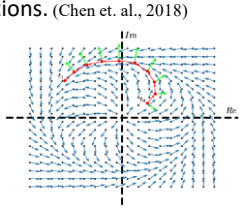


Backward Operation:

Adjoint sensitivity Method

$$\frac{dL}{d\alpha} = \int_{2\pi}^{0} \overline{\left(\frac{\partial L}{\partial \gamma_t}\right)} \frac{\partial f(\gamma_t, \alpha, t)}{\partial \alpha}\, dt$$

Where "$\overline{\ }$" is conjugate



Proposed Method : Hopf Activation

Hopf Activation

$\gamma_{t_j} \in \mathbb{C}^n$

$t_0, t_1 \in \mathbb{R}$

Final State $\gamma \in \mathbb{C}^n$

Initial State $\gamma \in \mathbb{C}^n$

Input constants $\alpha, \beta \in \mathbb{C}^n$

Evolution Function $f \in \mathbb{C}$

$$\gamma_{t_1} = \eta(\gamma_{t_0}, \alpha, \beta) = \gamma_{t_0} + \int_0^{2\pi} f(\gamma, t)\, dt \qquad f = (\alpha + \beta|\gamma|^2)\gamma$$

Hopf Activation Requirements

$\alpha = (a + ib) \qquad \beta = (c + id)$

1) Ensure Stability — Require: $c < 0$

2) Ensure complementary phases — Require: $bd > 0$

3) Solution Method — Numerical Methods give "approximate" solutions. (Chen et. al., 2018)
Adjoint sensitivity method calculates gradients from time t_1 to t_0.

Proposed Method : RNN Architecture and Control Methods

Input Size: n Units: u

Weights: $A, B \in \mathbb{C}^{(u \times u)}$

Input: $x_\tau \in \mathbb{R}^n$

DFT Frequency: $s = \begin{cases} 2(u+1) & \text{if } u > (n/2+1) \\ n+2 & \text{otherwise} \end{cases}$

State: $z_\tau''' \in \mathbb{C}^{(u+1)}$

$\frac{1}{\sqrt{s}}\mathcal{DFT}(x_\tau)$

Split: $z_\tau' = \quad = z_\tau''$

$x_\tau' = \quad = x_\tau''$

Maps: $\frac{1}{2}(z_\tau' + x_\tau') = \quad = z_\tau'' A + x_\tau'' B = z_{\tau+1}''$

$= \frac{z_\tau'' + z_{\tau+1}''}{\|z_\tau'' + z_{\tau+1}''\|}$

New State: $z_{\tau+1}'''$ Concatenate

$= \sigma(z_{\tau+1}'')$

$\mathcal{DFT}^{-1}(y_{\tau+1}'''\sqrt{s}) \rightarrow \quad = y_{\tau+1} \in \mathbb{R}^n$ Output

Control Methods

Method 1:

$$m_1^{\pm} = (\pm 1 + i0)$$

Method 2:

$$m_2 = -\left((z_{\tau+1}'')^{\frac{1}{u}}\right)^*$$

Where "*" is conjugate

m_1^+ m_1^-, m_2

$$\gamma_{t_1} = \eta(\gamma_{t_0}, \alpha, \beta)$$

Experimental - Setup

- Introduction
- Related Work
- Motivations
- Proposed Method
- **Experiments**
  - **Setup**
  - Results
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Experiments: Setup - Mackey Glass Dataset

Mackey Glass (Gallichio & Macheli, 2019)

Time-series data set testing a networks ability to predict chaotic dynamical systems.

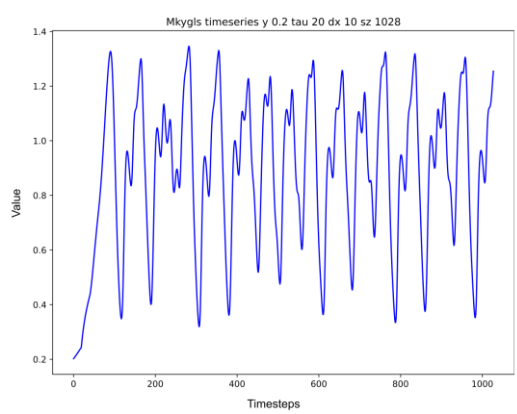
Total:	35,000	Windows
Train:	7,500	
Validation:	2,500	(25% split)
Test:	25,000	

Windows size: $n = 32 \,\&\, 64$

Recurrent Unit Sizes: $u = 8, 16, \& \, 24$

Root Mean Squared Error (RMSE)

$$RMSE = \sqrt{\frac{1}{S}\sum_{i=1}^{S}(y_i - \hat{y}_i)^2}$$



$$\dot{y}_t = \frac{0.2y_{t-\tau}}{\delta_x\left(1 + y_{t-\tau}^{\delta_x}\right)} - 0.1y_t \qquad \tau = 30 \quad y_0 = 0.2 \quad \delta_x = 10$$

Experiments: Setup – Copy Memory Dataset

Copy Memory

Network stress test for storing fixed sized data over long intervals. (Arjovsky et al. 2016); (S. Wisdom et at., 2014)

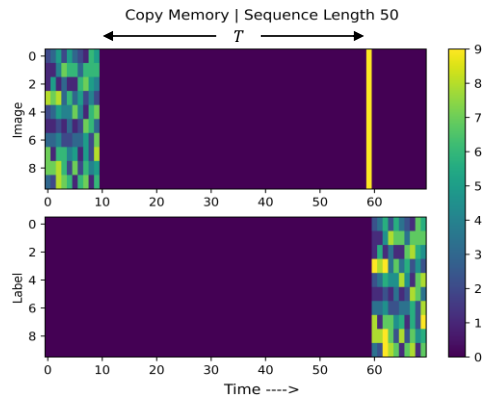
Total:	125,000	Examples
Training:	75,000	
Validation:	25,000	(25% split)
Testing:	25,000	

Sequences Length: $T = 200 \,\&\, 500$

Recurrent Unit Sizes: $u = 30, 60, \,\&\, 90$

Categorical Cross Entropy (CCE) Loss

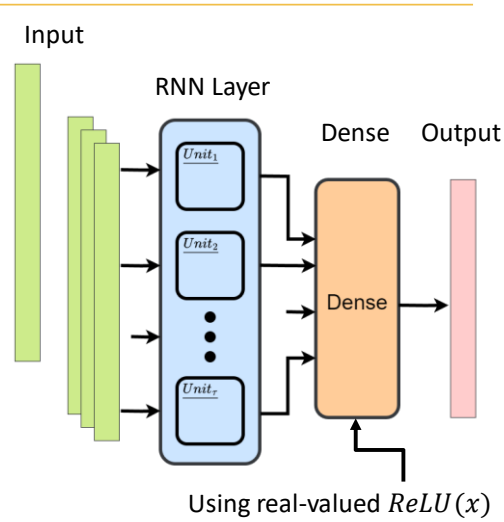
$$CCE = -\sum_{i=1}^{C} y_i \log(\hat{y}_i)$$



Cross Entropy Baseline (CEB): $CEB = \frac{10log(8)}{T + 20}$

Experiments: Setup – Training & Model Topology

Optimizer:	RMSProp
Momentum:	None
Discount Factor:	$\rho = 0.9$
Learning Rate:	0.001
Batch Size:	25
Early Stopping:	10 epochs



Experiments: Setup – Experiments

Parameter

Configuration (1)	$\eta(z''_{\tau+1}, m_1^+, m_1^-)$
Configuration (2)	$\eta(m_1^+, z''_{\tau+1}, m_1^-)$
Configuration (3)	$\eta(m_1^+, m_1^+, m_2)$
Configuration (4)	$\eta(m_1^+, z''_{\tau+1}, m_2)$
Configuration (5)	$\eta(z''_{\tau+1}, m_1^+, m_2)$
Configuration (6)	$\eta(z''_{\tau}, z''_{\tau+1}, m_1^-)$
Configuration (7)	$\eta(z''_{\tau}, z''_{\tau+1}, m_2)$

Goal: Evaluate different activation configurations

Performance

(mr): $modReLU(z) = \begin{cases} (|z| + b)\frac{z}{|z|} & \text{if } |z| + b \geq 0 \\ 0 & \text{if } |z| + b < 0 \end{cases}$

(sl): $SigLog(z) = \frac{z}{c + |z|/r}$

(cc): $CpxCard(z) = \frac{1}{2}(1 + \cos(\measuredangle z))\, z$

(t): $tanh(z) = \frac{e^z + e^{-z}}{e^z - e^{-z}}$

(sr): $splitReLU(z) = ReLU(a) + iReLU(b) \quad z = (a + ib)$

Goal: Evaluate 2 best parameter configurations against well established activations.

Unit size: u $\quad m_1^{\pm} = (\pm 1 + i0) \quad m_2 = -\left((z''_{\tau+1})^{\frac{1}{u}}\right)^*$

A single matrix initialization is shared for each configuration or activation set for both experiments.

Experiments - Results

- Introduction
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  - **Results**
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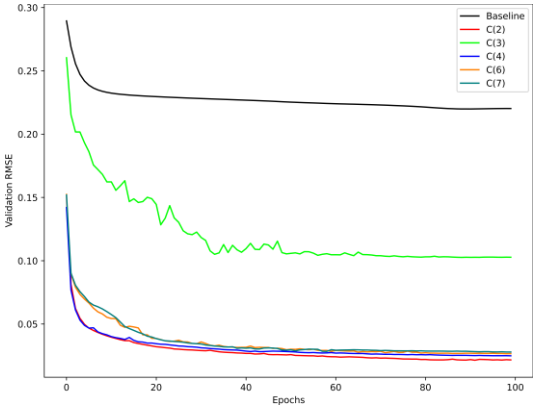
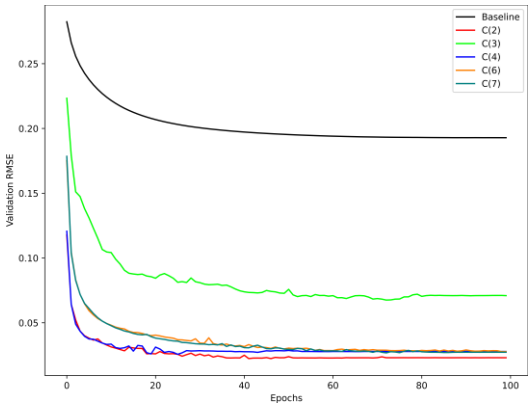
Experiments: Results | Parameter | Mackey Glass | Training

Average Validation RMSE Loss – Window Size 32



Model set with 16 recurrent units
Loss Curves represent the average of 26 experimental runs

Average Validation RMSE Loss – Window Size 64



All configurations achieve a lower validation RMSE than the Baseline model.

Experiments: Results | Parameter | Mackey Glass | Evaluation

Configurations C(2) and C(4) performed best

Significant improvements for C(2) and C(4) as window sizes increased from 32 to 64.

$$\%\ Imprv = \frac{(A - B)}{A} \times 100$$

Average RMSE Evaluation Results for Units 8-24

Window Size		32		64	
Model		RMSE		RMSE	
Config	Units 8-24	Valid	Eval	Valid	RMSE
Base	**Average**	0.2065	0.1972	0.1898	0.1800
C(2)	**Average**	0.0360	0.0249	0.0369	0.0241
C(3)	**Average**	0.1176	0.0810	0.0920	0.0703
C(4)	**Average**	0.0367	0.0255	0.0382	0.0241
C(6)	**Average**	0.0380	0.0256	0.0527	0.0364
C(7)	**Average**	0.0372	0.0253	0.0534	0.0361

Average RMSE % Improvement | Window Size 32

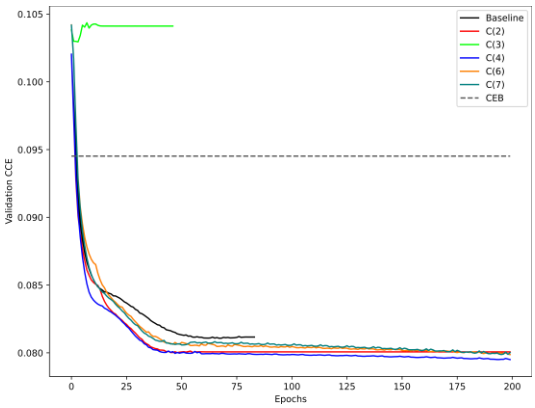
Reference	Base	C(2)	C(3)	C(4)	C(6)	C(7)
Base		87.4%	59.0%	87.1%	87.0%	87.2%
C(2)	-692.0%		-225.1%	-2.3%	-2.9%	-1.6%
C(3)	-143.6%	69.2%		68.5%	68.3%	68.7%
C(4)	-674.3%	2.2%	-217.8%		-0.6%	0.7%
C(6)	-669.6%	2.8%	-215.9%	0.6%		1.3%
C(7)	-679.5%	1.6%	-220.0%	-0.7%	-1.3%	
Ave:	**-571.82%**	**32.65%**	**-163.97%**	**30.65%**	**30.11%**	**31.25%**

Average RMSE % Improvement | Window Size 64

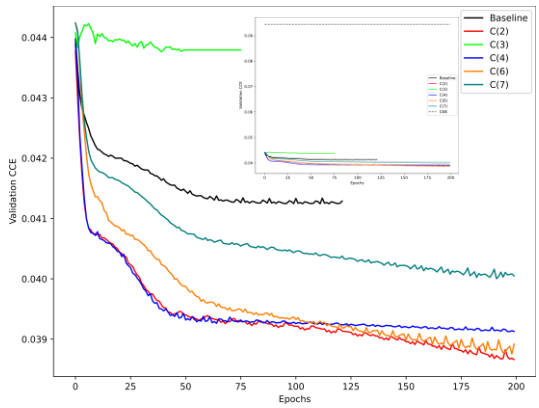
Reference	Base	C(2)	C(3)	C(4)	C(6)	C(7)
Base		86.6%	60.9%	86.6%	79.8%	79.9%
C(2)	-645.4%		-191.3%	0.1%	-50.7%	-49.7%
C(3)	-155.9%	65.7%		65.7%	48.3%	48.6%
C(4)	-646.1%	-0.1%	-191.6%		-50.8%	-49.9%
C(6)	-394.7%	33.6%	-93.3%	33.7%		0.6%
C(7)	-397.8%	33.2%	-94.5%	33.3%	-0.6%	
Ave:	**-448.00%**	**43.80%**	**-101.96%**	**43.87%**	**5.19%**	**5.90%**

Experiments: Results | Parameter | Copy Memory | Training

Average Validation CCE Loss – Sequence Length 200



Average Validation CCE Loss – Sequence Length 500



Model set with 60 recurrent units
Loss Curves represent the average of 26 experimental runs

Configurations stopping short of 200 epochs is due to early stopping.

Experiments: Results | Parameter | Copy Memory | Evaluation

Results indicate little to no effect from the activation function with respect to Base model.

Results inconclusive.

$$\%\ Imprv = \frac{(A - B)}{A} \times 100$$

Average CCE Evaluation Results for Units 30-90

Sequence Length		200		500	
Model		CCE		CCE	
Config	Units 30-90	Valid	Eval	Valid	Eval
Base	**Average**	0.0843	0.0819	0.0410	0.0405
C(2)	**Average**	0.0827	0.0799	0.0401	0.0393
C(3)	**Average**	0.1024	0.1009	0.0441	0.0438
C(4)	**Average**	0.0829	0.0801	0.0401	0.0394
C(6)	**Average**	0.0835	0.0804	0.0404	0.0397
C(7)	**Average**	0.0834	0.0805	0.0406	0.0399

Average CCE % Improvement | Sequence Length 200

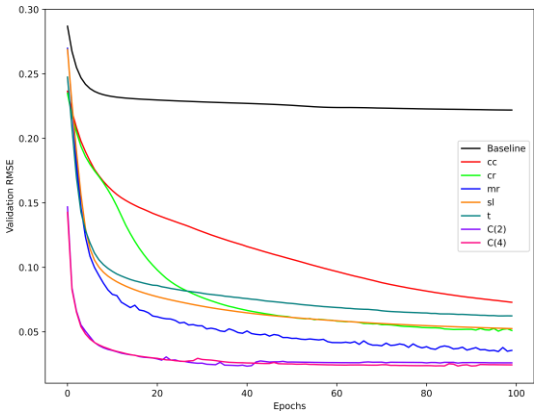
Reference	Base	C(2)	C(3)	C(4)	C(6)	C(7)
Base		2.5%	-23.2%	2.3%	1.9%	1.8%
C(2)	-2.6%		-26.3%	-0.2%	-0.7%	-0.7%
C(3)	18.8%	20.8%		20.7%	20.3%	20.3%
C(4)	-2.4%	0.2%	-26.1%		-0.4%	-0.5%
C(6)	-1.9%	0.7%	-25.5%	0.4%		-0.1%
C(7)	-1.8%	0.7%	-25.4%	0.5%	0.1%	
Ave:	**2.03%**	**4.99%**	**-25.30%**	**4.74%**	**4.23%**	**4.16%**

Average CCE % Improvement | Sequence Length 500

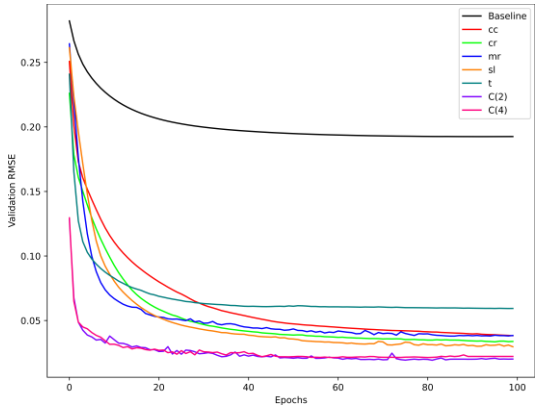
Reference	Base	C(2)	C(3)	C(4)	C(6)	C(7)
Base		2.8%	-8.3%	2.7%	2.0%	1.4%
C(2)	-2.8%		-11.4%	0.0%	-0.8%	-1.4%
C(3)	7.7%	10.2%		10.2%	9.5%	9.0%
C(4)	-2.8%	0.0%	-11.3%		-0.8%	-1.4%
C(6)	-2.0%	0.8%	-10.5%	0.8%		-0.6%
C(7)	-1.4%	1.4%	-9.8%	1.3%	0.6%	
Ave:	**-0.29%**	**3.04%**	**-10.27%**	**2.99%**	**2.10%**	**1.41%**

Experiments: Results | Performance | Mackey Glass | Training

Average Validation RMSE Loss – Window Size 32



Average Validation RMSE Loss – Window Size 64



Model set with 16 recurrent units
Loss Curves represent the average of 26 experimental runs

Configurations (2) and (4) reached lowest RMSE values compare to other activation.

Experiments: Results | Performance | Mackey Glass | Evaluation

Configurations C(2) and C(4) were the best performing.

Configurations C(2) and C(4) performed 40% - 50% better than other activations.

$$\%\ Imprv = \frac{(A - B)}{A} \times 100$$

Average RMSE Evaluation Results for Units 8-24

Window Size		32		64	
Model		RMSE		RMSE	
Activation	Units 8-24	Valid	Eval	Valid	RMSE
CpxCard	**Average**	0.0994	0.0681	0.0711	0.0480
cpxReLU	**Average**	0.0768	0.0506	0.0682	0.0431
modReLU	**Average**	0.0732	0.0437	0.0546	0.0351
SigLog	**Average**	0.0791	0.0614	0.0577	0.0348
tanh	**Average**	0.0751	0.0587	0.0700	0.0547
Base	**Average**	0.2058	0.1967	0.1897	0.1800
C(2)	**Average**	0.0367	0.0243	0.0362	0.0227
C(4)	**Average**	0.0363	0.0246	0.0373	0.0242

Average RMSE % Improvement | Window Size 32

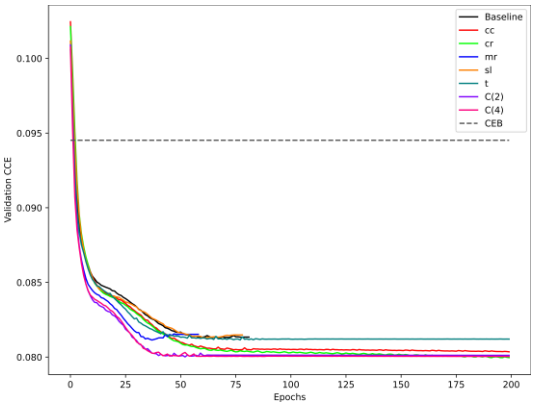
Reference	CpxCard	cpxReLU	modReLU	SigLog	*tanh*	C(2)	C(4)
CpxCard		10.1%	26.8%	27.5%	-14.0%	52.8%	49.7%
cpxReLU	-11.3%		18.5%	19.3%	-26.9%	47.4%	44.0%
modReLU	-36.5%	-22.7%		1.0%	-55.7%	35.5%	31.3%
SigLog	-37.9%	-23.9%	-1.0%		-57.2%	34.9%	30.6%
tanh	12.3%	21.2%	35.8%	36.4%		58.6%	55.9%
Ave:	-30.0%	-15.4%	8.6%	9.6%	-50.3%	46.1%	41.6%

Average RMSE % Improvement | Window Size 64

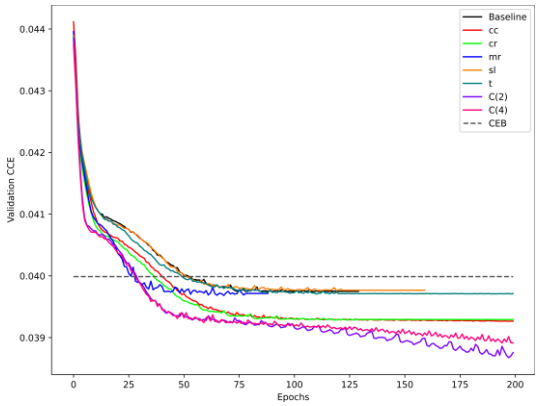
Reference	CpxCard	cpxReLU	modReLU	SigLog	*tanh*	C(2)	C(4)
CpxCard		25.7%	35.8%	9.8%	13.9%	64.4%	63.9%
cpxReLU	-34.7%		13.6%	-21.4%	-15.9%	52.1%	51.4%
modReLU	-55.8%	-15.7%		-40.4%	-34.1%	44.5%	43.8%
SigLog	-10.9%	17.6%	28.8%		4.5%	60.5%	60.0%
tanh	-16.2%	13.7%	25.4%	-4.7%		58.6%	58.1%
Ave:	-58.6%	-14.1%	3.3%	-41.6%	-34.6%	52.7%	51.9%

Experiments: Results | Performance | Copy Memory | Training

Average Validation CCE Loss – Sequence Length 200



Average Validation CCE Loss – Sequence Length 500



Model set with 60 recurrent units
Loss Curves represent the average of 26 experimental runs

Configurations stopping short of 200 epochs is due to early stopping.

Experiments: Results | Performance | Copy Memory | Evaluation

Results indicate little to no effect from the Activation function with respect to Base model.

Results inconclusive.

$$\% \, Imprv = \frac{(A - B)}{A} \times 100$$

Average CCE Evaluation Results for Units 30-90

Sequence Length		200		500	
Model		CCE		CCE	
Activation	Units 30-90	Valid	Eval	Valid	Eval
CpxCard	**Average**	0.0836	0.0811	0.0405	0.0399
cpxReLU	**Average**	0.0836	0.0811	0.0402	0.0396
modReLU	**Average**	0.0839	0.0813	0.0408	0.0402
SigLog	**Average**	0.0844	0.0817	0.0408	0.0402
tanh	**Average**	0.0836	0.0813	0.0407	0.0402
Base	**Average**	0.0841	0.0817	0.0408	0.0403
C(2)	**Average**	0.0826	0.0798	0.0401	0.0393
C(4)	**Average**	0.0827	0.0799	0.0401	0.0394

Average CCE % Improvement | Sequence Length 200

Reference	CpxCard	cpxReLU	modReLU	SigLog	*tanh*	C(2)	C(4)
CpxCard		-0.03%	-0.25%	-0.79%	-0.26%	1.57%	1.41%
cpxReLU	0.03%		-0.23%	-0.77%	-0.23%	1.59%	1.43%
modReLU	0.25%	0.23%		-0.54%	-0.01%	1.82%	1.66%
SigLog	0.79%	0.76%	0.53%		0.53%	2.34%	2.18%
tanh	0.26%	0.23%	0.01%	-0.53%		1.82%	1.66%
Ave:	-0.13%	-0.16%	-0.42%	-1.04%	-0.43%	1.66%	1.48%

Average CCE % Improvement | Sequence Length 500

Reference	CpxCard	cpxReLU	modReLU	SigLog	*tanh*	C(2)	C(4)
CpxCard		0.85%	-0.71%	-0.87%	-0.75%	1.60%	1.38%
cpxReLU	-0.86%		-1.57%	-1.74%	-1.61%	0.75%	0.53%
modReLU	0.70%	1.55%		-0.16%	-0.04%	2.29%	2.07%
SigLog	0.86%	1.71%	0.16%		0.12%	2.45%	2.23%
tanh	0.74%	1.59%	0.04%	-0.13%		2.33%	2.11%
Ave:	-0.09%	0.88%	-0.90%	-1.09%	-0.95%	1.74%	1.48%

Discussion

- Introduction
- Related Work
- Motivations
- Proposed Method
- Experiments
  - Setup
  - Results
- **Discussion**
- Conclusion
- Future Works

Discussion: Configurations

$m_1^{\pm} = (\pm 1 + i0)$ $\quad m_2 = -\left((z''_{\tau+1})^{\frac{1}{u}}\right)^*$

Configurations (1) & (5) failed training

C(1) = $\eta(z''_{\tau+1}, m_1^+, m_1^-)$

C(5) = $\eta(z''_{\tau+1}, m_1^+, m_2)$

Configurations (3) worst performing

C(3) = $\eta(m_1^+, m_1^+, m_2)$

Configurations (6) & (7) perform well

C(6) = $\eta(z''_{\tau}, z''_{\tau+1}, m_1^-)$

C(7) = $\eta(z''_{\tau}, z''_{\tau+1}, m_2)$

Configurations (2) & (4) perform best

C(2) = $\eta(m_1^+, z''_{\tau+1}, m_1^-)$

C(4) = $\eta(m_1^+, z''_{\tau+1}, m_2)$

Discussion: Performance

Baseline model & $tanh$ were the worst upon both datasets.

CpxCard & cpxReLU performed poor on Mackey Glass.

modReLU & SigLog performed poor on Copy Memory.

Configurations (2) & (4) perform best on all datasets.

Common types of activation functions performed in a similar fashions on similar datasets.

Conclusion

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Conclusion

The Hopf (limit cycle) Activation can be used as a non-linear activation function when the restrictions of $c < 0$ and $bd > 0$ are maintained, where $\alpha = (a + ib)$ & $\beta = (c + id)$.

The configurations where the state z''_{t+1} is in the α position performs the best against the Mackey Glass and Copy Memory datasets. The β position value was not effective / detectable.

The Hopf activation performs better than or equal to the currently provided activation functions.

Future Work

- Introduction
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- Motivations
- Proposed Method
- Experiments
  - Setup
  - Results
- Discussion
- Conclusion
- **Future Works**

Future Works

Experiment with more established RNN architectures.

Transition from a purely complex domain to a sympletic domain to learn non-cyclic shapes.

$$\dot{x} = (ax - by)\mu + (ex - fy)(x^2 + y^2)$$
$$\dot{y} = (cx + dy)\mu + (gx + hy)(x^2 + y^2)$$

$$\alpha = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \qquad \beta = \begin{bmatrix} e & f \\ g & h \end{bmatrix}$$

Questions?

Thank You!

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References

1. E. Kanjo, E. M. G. Younis, and C. S. Ang, "Deep learning analysis of mobile physiological, environmental and location sensor data for emotion detection," *Information Fusion*, vol. 49, pp. 46–56, Sep. 2019, doi: 10.1016/j.inffus.2018.09.001.
2. A. Moghar and M. Hamiche, "Stock Market Prediction Using LSTM Recurrent Neural Network," *Procedia Computer Science*, vol. 170, pp. 1168–1173, Jan. 2020, doi: 10.1016/j.procs.2020.03.049.
3. H. Yan and H. Ouyang, "Financial Time Series Prediction Based on Deep Learning," *Wireless Pers Commun*, vol. 102, no. 2, pp. 683–700, Sep. 2018, doi: 10.1007/s11277-017-5086-2.
4. Z. Luo and M. Zhu, "Recurrent Neural Networks with Mixed Hierarchical Structures for Natural Language Processing," in *2021 International Joint Conference on Neural Networks (IJCNN)*, Jul. 2021, pp. 1–8. doi: 10.1109/IJCNN52387.2021.9533347.
5. I. Orlovskyi, "Mathematical model of an electromechanical object synthesized as modified recurrent neural networks," in *2017 International Conference on Modern Electrical and Energy Systems (MEES)*, Nov. 2017, pp. 292–295. doi: 10.1109/MEES.2017.8248914.
6. Q. Xiao, X. Chang, X. Zhang, and X. Liu, "Multi-Information Spatial–Temporal LSTM Fusion Continuous Sign Language Neural Machine Translation," *IEEE Access*, vol. 8, pp. 216718–216728, 2020, doi: 10.1109/ACCESS.2020.3039539.
7. Z. Lei, S. Gao, H. Hasegawa, Z. Zhang, M. Zhou, and K. Sedraoui, "Fully Complex-Valued Gated Recurrent Neural Network for Ultrasound Imaging," *IEEE Trans. Neural Netw. Learning Syst.*, pp. 1–14, 2023, doi: 10.1109/TNNLS.2023.3282231.

References

8. Z. Zhang, Z. Lei, M. Zhou, H. Hasegawa, and S. Gao, "Complex-Valued Convolutional Gated Recurrent Neural Network for Ultrasound Beamforming," *IEEE Transactions on Neural Networks and Learning Systems*, pp. 1–12, 2024, doi: 10.1109/TNNLS.2024.3384314.
9. Y. Ren, W. Jiang, and Y. Liu, "Complex-valued Parallel Convolutional Recurrent Neural Networks for Automatic Modulation Classification," in *2022 IEEE 25th International Conference on Computer Supported Cooperative Work in Design (CSCWD)*, May 2022, pp. 804–809. doi: 10.1109/CSCWD54268.2022.9776161.
10. W. Xie, G. Ma, F. Zhao, H. Liu, and L. Zhang, "PolSAR image classification via a novel semi-supervised recurrent complex-valued convolution neural network," *Neurocomputing*, vol. 388, pp. 255–268, May 2020, doi: 10.1016/j.neucom.2020.01.020.
11. R. Zhang, P.-Y. Zhang, M.-R. Gao, L.-H. Pan, X.-L. Bai, and N. Zhao, "Self-optimizing multi-domain auxiliary fusion deep complex convolution recurrent network for speech enhancement," *Digital Signal Processing*, vol. 134, p. 103897, Apr. 2023, doi: 10.1016/j.dsp.2022.103897.
12. S. Hochreiter and J. Schmidhuber, "Long Short-Term Memory," *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, Nov. 1997, doi: 10.1162/neco.1997.9.8.1735.
13. K. Cho, B. van Merrienboer, D. Bahdanau, and Y. Bengio, "On the Properties of Neural Machine Translation: Encoder-Decoder Approaches," *arXiv:1409.1259 [cs, stat]*, Oct. 2014, Accessed: Mar. 10, 2022. [Online]. Available: http://arxiv.org/abs/1409.1259
14. S. Chandar, C. Sankar, E. Vorontsov, S. E. Kahou, and Y. Bengio, "Towards Non-Saturating Recurrent Units for Modelling Long-Term Dependencies," *AAAI*, vol. 33, pp. 3280–3287, Jul. 2019, doi: 10.1609/aaai.v33i01.33013280.

References

15. G. Van Houdt, C. Mosquera, and G. Nápoles, "A review on the long short-term memory model," *Artif Intell Rev*, vol. 53, no. 8, pp. 5929–5955, Dec. 2020, doi: 10.1007/s10462-020-09838-1.
16. M. Arjovsky, A. Shah, and Y. Bengio, "Unitary evolution recurrent neural networks," in *Proceedings of the 33rd International Conference on International Conference on Machine Learning - Volume 48*, in ICML'16. New York, NY, USA: JMLR.org, Jun. 2016, pp. 1120–1128.
17. S. Wisdom, T. Powers, J. Hershey, J. Le Roux, and L. Atlas, "Full-Capacity Unitary Recurrent Neural Networks," in *Advances in Neural Information Processing Systems*, Curran Associates, Inc., 2016, pp. 4887–4895. Accessed: Mar. 20, 2024. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2016/hash/d9ff90f4000eacd3a6c9cb27f78994cf-Abstract.html
18. M. Wolter and A. Yao, "Complex Gated Recurrent Neural Networks," in *Advances in Neural Information Processing Systems*, Curran Associates, Inc., 2018, pp. 10557–10567. Accessed: Jan. 29, 2024. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2018/hash/652cf38361a209088302ba2b8b7f51e0-Abstract.html
19. Z. Lei, S. Gao, H. Hasegawa, Z. Zhang, M. Zhou, and K. Sedraoui, "Fully Complex-Valued Gated Recurrent Neural Network for Ultrasound Imaging," *IEEE Trans. Neural Netw. Learning Syst.*, pp. 1–14, 2023, doi: 10.1109/TNNLS.2023.3282231.
20. J. Zhang, Y. Lin, Z. Song, and I. Dhillon, "Learning Long Term Dependencies via Fourier Recurrent Units," in *Proceedings of the 35th International Conference on Machine Learning*, PMLR, Jul. 2018, pp. 5815–5823. Accessed: Jun. 24, 2022. [Online]. Available: https://proceedings.mlr.press/v80/zhang18h.html
21. A. Orvieto *et al.*, "Resurrecting Recurrent Neural Networks for Long Sequences," in *Proceedings of the 40th International Conference on Machine Learning*, PMLR, Jul. 2023, pp. 26670–26698. Accessed: Apr. 14, 2024. [Online]. Available: https://proceedings.mlr.press/v202/orvieto23a.html

References

22. T. L. Clarke, "Generalization of neural networks to the complex plane," in *1990 IJCNN International Joint Conference on Neural Networks*, Jun. 1990, pp. 435–440 vol.2. doi: 10.1109/IJCNN.1990.137751.
23. T. Kim and T. Adalı, "Approximation by Fully Complex Multilayer Perceptrons," *Neural Computation*, vol. 15, no. 7, pp. 1641–1666, Jul. 2003, doi: 10.1162/089976603321891846.
24. C. Trabelsi *et al.*, "Deep Complex Networks," presented at the International Conference on Learning Representations, Feb. 2018, p. NA. Accessed: Mar. 21, 2024. [Online]. Available: https://openreview.net/forum?id=H1T2hmZAb
25. T. Nitta, "An Extension of the Back-Propagation Algorithm to Complex Numbers," *Neural Networks*, vol. 10, no. 8, pp. 1391–1415, Nov. 1997, doi: 10.1016/S0893-6080(97)00036-1.
26. C.-A. Popa, "Complex-valued convolutional neural networks for real-valued image classification," in *2017 International Joint Conference on Neural Networks (IJCNN)*, May 2017, pp. 816–822. doi: 10.1109/IJCNN.2017.7965936.
27. G. M. Georgiou and C. Koutsougeras, "Complex domain backpropagation," *IEEE Trans. Circuits Syst. II*, vol. 39, no. 5, pp. 330–334, May 1992, doi: 10.1109/82.142037.
28. P. Virtue, S. X. Yu, and M. Lustig, "Better than real: Complex-valued neural nets for MRI fingerprinting," in *2017 IEEE International Conference on Image Processing (ICIP)*, Beijing: IEEE, Sep. 2017, pp. 3953–3957. doi: 10.1109/ICIP.2017.8297024.
29. E. M. Izhikevich, *Dynamical Systems in Neuroscience: The Geometry of Excitability and Bursting*. in Computational Neuroscience Series. Cambridge, MA, USA: MIT Press, 2006.

References

30. S. Wiggins, *Introduction to Applied Nonlinear Dynamical Systems and Chaos*, vol. 2. in Texts in Applied Mathematics, vol. 2. New York, NY: Springer, 1990. doi: 10.1007/978-1-4757-4067-7.
31. R. T. Q. Chen, Y. Rubanova, J. Bettencourt, and D. K. Duvenaud, "Neural Ordinary Differential Equations," in *Advances in Neural Information Processing Systems*, Curran Associates, Inc., 2018, pp. 6572–6583. Accessed: Mar. 25, 2024. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2018/hash/69386f6bb1dfed68692a24c8686939b9-Abstract.html
32. C. Gallicchio and A. Micheli, "Reservoir Topology in Deep Echo State Networks," in *Artificial Neural Networks and Machine Learning – ICANN 2019: Workshop and Special Sessions*, I. V. Tetko, V. Kůrková, P. Karpov, and F. Theis, Eds., Cham: Springer International Publishing, 2019, pp. 62–75. doi: 10.1007/978-3-030-30493-5_6.

Proposed Method : RNN Layer

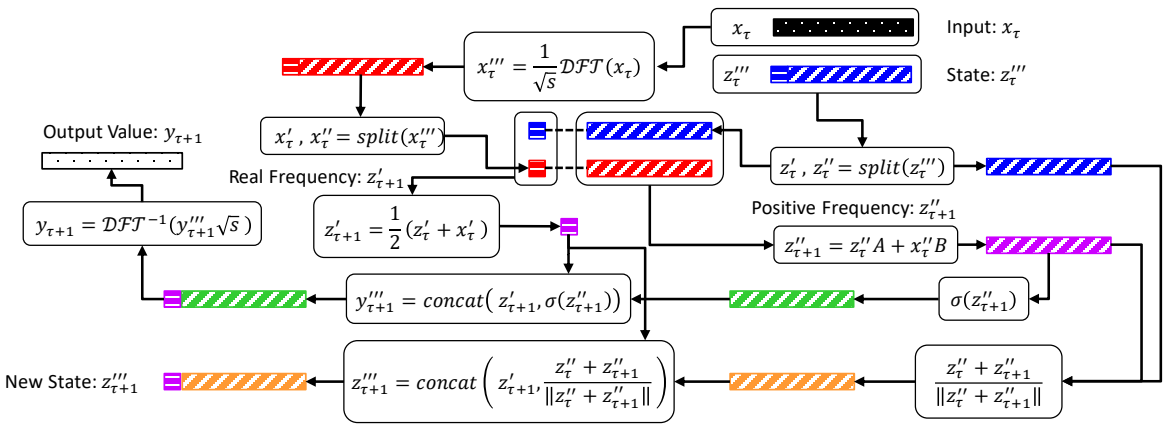
Input Size: n

State: $x_\tau \in \mathbb{R}^n$

Recurrent Units: u

State: $z_\tau''' \in \mathbb{C}^{(u+1)}$

DFT Sample Frequency: $s = \begin{cases} 2(u+1) & \text{if } u > (n/2+1) \\ n+2 & \text{otherwise} \end{cases}$



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